HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

# Electronic Circuit Design

Slides to the lecture

## Chapter 9: Analog-to-Digital Conversion

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Winter semester 2026/27

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# 9 Analog-to-Digital Conversion

# 9.1 Sampling and Aliasing

## The sampling theorem

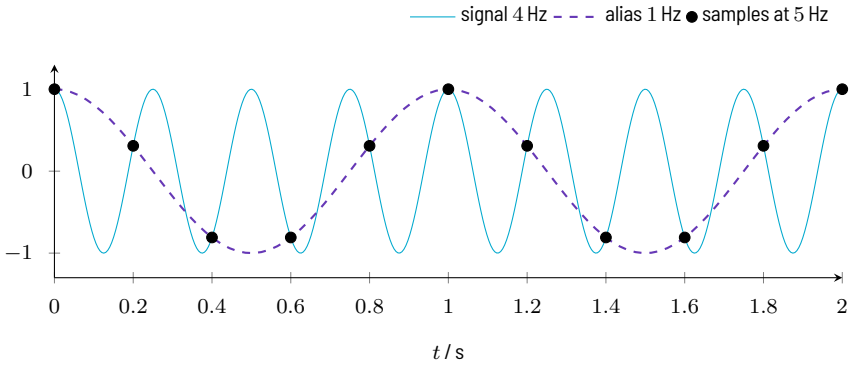
- a signal is sampled at the instants $t_k = k\,T_s$ with the sampling frequency $f_s = 1/T_s$

**Sampling theorem (Nyquist–Shannon)**

A signal that contains no frequencies at or above $f_s/2$ is completely determined by its samples taken at the rate $f_s$. $f_s/2$ is called the **Nyquist frequency**.

- the condition is $f_{\text{max}} < f_s/2$, i.e. more than two samples per period of the fastest component
- it applies to **every** component of the input, also to noise and interference
- capstone: $f_s = 5$ Hz $\Rightarrow$ only components below $2.5$ Hz are allowed

# Aliasing

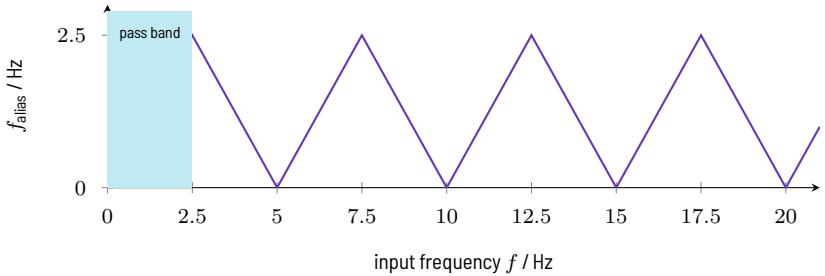



- a $4$ Hz cosine sampled at $5$ Hz gives the same samples as a $1$ Hz cosine

## Alias frequency

$f_{\text{alias}} = |f - k\, f_s|$, with the integer $k$ that makes $f_{\text{alias}} \leq f_s/2$

# Folding




- every frequency is folded into the range $0 \ldots f_s/2$
- $f_s = 5$ Hz: $4$ Hz $\rightarrow 1$ Hz; $12$ Hz $\rightarrow 2$ Hz; $49$ Hz $\rightarrow 1$ Hz; $50$ Hz $\rightarrow 0$ Hz
- after sampling an alias cannot be distinguished from a real signal component

## The 50 Hz problem

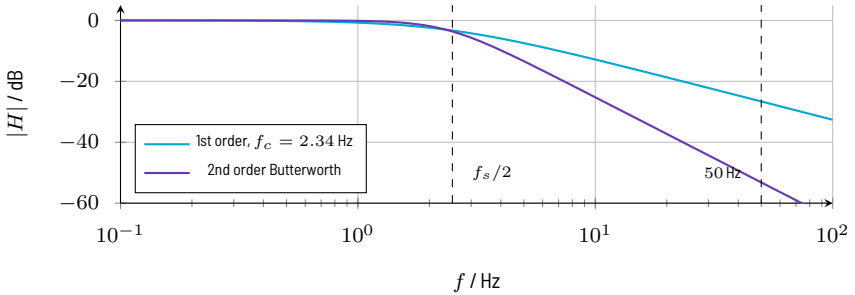
- mains interference ($50$ Hz) is coupled into every sensor cable
- capstone: $f_s = 5\,\text{Hz} \Rightarrow f_{\text{alias}} = |50\,\text{Hz} - 10 \cdot 5\,\text{Hz}| = 0\,\text{Hz}$
  - every sample sees the 50 Hz sine at the same phase
  - the result is a **constant DC offset** that biases every reading
  - its size depends on the (random) phase: anywhere between $-\hat{V}$ and $+\hat{V}$
  - nothing on the display looks wrong: the error is invisible
- with $f_s = 4.9\,\text{Hz}$ the alias is $|50\,\text{Hz} - 10 \cdot 4.9\,\text{Hz}| = 1\,\text{Hz}$: a slow beat
- **the anti-aliasing low-pass in front of the S&H stage is mandatory**

# Falstad: sampling 50 Hz hum

- input: $2$ V DC (the wanted signal) plus $0.5$ V at $50$ Hz
- two S&H stages ($R = 1\,\text{k}\Omega$, $C_H = 100$ nF, track window $1$ ms)
  - $f_s = 5$ Hz: every sample is $2.14$ V: an offset of $0.14$ V
  - $f_s = 4.9$ Hz: $2.14$ V, $2.50$ V, $2.14$ V, $1.58$ V, ...: a $1$ Hz sine
- change the phase of the $50$ Hz source: the offset at $5$ Hz changes between $1.5$ V and $2.5$ V

▶ Falstad: `examples/adc/alias50.txt`

# Anti-aliasing filter




- pass the wanted signal, attenuate everything above $f_s/2$ below the resolution
- at $50$ Hz: 1st order $-26.6$ dB (factor 21), 2nd order $-53$ dB (factor 460)
- the filter must be **before** the S&H: after sampling the damage is done

## Design card: Nyquist & Anti-aliasing

- sampling theorem: all input components must be below $f_s/2$ (Nyquist frequency)
- alias: $f_{\text{alias}} = |f - kf_s|$, nearest multiple $kf_s$; result lies in $0 \ldots f_s/2$
- capstone: $f_s = 5$ Hz, $50$ Hz hum $\rightarrow |50 - 10 \cdot 5|$ Hz $= 0$ Hz: constant offset on every reading
- low-pass before the S&H: $f_c \approx f_s/2$ (capstone $2.5$ Hz), e.g. RC: $f_c = 1/(2\pi RC)$
- required attenuation: interference amplitude $\times |H(f)| < \frac{1}{2}$ LSB (better $\frac{1}{4}$ LSB)
- 1st order: $|H| \approx f_c/f$ ($-20$ dB/decade), 2nd order: $(f_c/f)^2$ ($-40$ dB/decade)
- higher $f_s$ (oversampling) relaxes the filter: more distance between signal band and $f_s/2$

# 9.2 Quantization

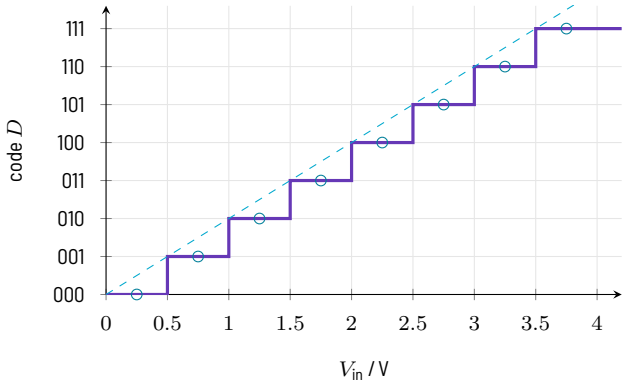
## Resolution and LSB

**LSB**

$$\text{LSB} = \frac{V_{\text{FS}}}{2^N} \qquad (V_{\text{FS}}\text{: full-scale range, } N\text{: number of bits})$$

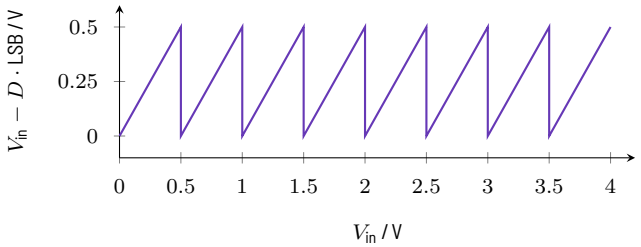
- capstone: $N = 3$, $V_{\text{FS}} = 4$ V: LSB $= 0.5$ V, 8 codes 000 …111
- 10 bit, $3.3$ V (microcontroller): $3.2$ mV; 16 bit, $5$ V: $76$ µV
- code for an input voltage (our flash converter):
  $D = \left\lfloor \frac{V_{\text{in}}}{\text{LSB}} \right\rfloor$, limited to $0 \ldots 2^N - 1$
- back to a voltage: $V_D = D \cdot \text{LSB}$ (lower edge) or $(D + \frac{1}{2})$ LSB (bin centre)

# Transfer characteristic




- staircase: thresholds at $k \cdot$ LSB ($k = 1 \ldots 7$), the code changes at $0.5$ V, $1.0$ V, ..., $3.5$ V
- circles: bin centres $0.25$ V, $0.75$ V, ..., $3.75$ V: safe test points

# Quantization error




- with thresholds at $k \cdot$ LSB: error $0 \ldots 1$ LSB (truncation), mean $\frac{1}{2}$ LSB
- shift all thresholds by $\frac{1}{2}$ LSB (ladder: $R/2$ at the bottom, $3R/2$ at the top): error $\pm\frac{1}{2}$ LSB
- or read the code as the bin centre $(D + \frac{1}{2})$ LSB: also $\pm\frac{1}{2}$ LSB

## Quantization noise and SNR

- for a busy signal the error behaves like noise, uniformly distributed over one LSB
  - rms value $V_{q,\text{rms}} = \dfrac{\text{LSB}}{\sqrt{12}}$
- full-scale sine: $V_{\text{rms}} = \dfrac{V_{\text{FS}}}{2\sqrt{2}} = \dfrac{2^N\,\text{LSB}}{2\sqrt{2}}$

**Signal-to-noise ratio of an ideal $N$-bit converter**

$$\text{SNR} = 20\log_{10}\frac{2^N\sqrt{12}}{2\sqrt{2}}\,\text{dB} = 6.02\,N + 1.76\,\text{dB}$$

- 3 bit: $19.8$ dB; 8 bit: $49.9$ dB; 12 bit: $74.0$ dB; 16 bit: $98.1$ dB
- every additional bit: $6$ dB more (factor 2 in amplitude)

## Falstad: quantization made visible

- 3-bit flash converter (next sections) followed by a 3-bit R-2R DAC
  - the DAC turns the code back into $V_q = D \cdot 0.5\,\text{V}$
- input: triangle $0\,\text{V} \ldots 4\,\text{V}$, $0.25\,\text{Hz}$
- the scope shows $V_{\text{in}}$ and the staircase $V_q$; the difference is always between $0\,\text{V}$ and $0.5\,\text{V}$
- example readings: $V_{\text{in}} = 1.8\,\text{V} \rightarrow V_q = 1.5\,\text{V}$; $V_{\text{in}} = 3.98\,\text{V} \rightarrow V_q = 3.5\,\text{V}$

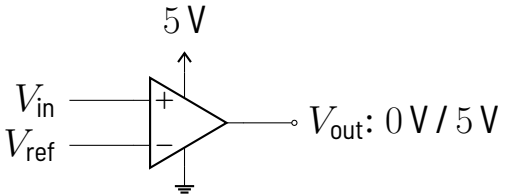
▶ Falstad: `examples/adc/quant_error.txt`

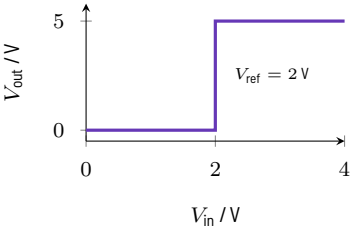
## Design card: Reference Ladder & Quantization

- LSB $= V_{\text{FS}}/2^N$; capstone: $4\,\text{V}/8 = 0.5\,\text{V}$
- ladder of $2^N$ equal resistors from $V_{\text{ref}} = V_{\text{FS}}$ to ground: taps $V_k = k \cdot V_{\text{ref}}/2^N$, $k = 1 \ldots 2^N - 1$
- capstone: $8 \times 1\,\text{k}\Omega$ from $4\,\text{V}$: taps $0.5\,\text{V}$, $1.0\,\text{V}$, ..., $3.5\,\text{V}$; $I = 0.5\,\text{mA}$
- code $D = \lfloor V_{\text{in}}/\text{LSB} \rfloor$; quantization error $0 \ldots 1$ LSB; with $R/2$ at the bottom and $3R/2$ at the top: $\pm\frac{1}{2}$ LSB
- test points: bin centres $(D + \frac{1}{2})$ LSB = $0.25\,\text{V}$, $0.75\,\text{V}$, ..., $3.75\,\text{V}$
- ladder current $\gg$ comparator input currents; tap resistance $\leq R_{\text{total}}/4 = 2\,\text{k}\Omega$
- reference: stiff voltage source (reference IC + buffer); its error scales all thresholds
- SNR of an ideal converter: $6.02\,N + 1.76\,\text{dB}$

# 9.3 The Comparator

## Open-loop op-amp as comparator




- $V_{out} = V_{OH}$ if $V_{in} > V_{ref}$, otherwise $V_{OL}$
- single $5$ V supply: output levels fit 5 V logic directly (rail-to-rail output)
- Falstad: op-amp with **max output** $5$ V, **min output** $0$ V

## Real comparators

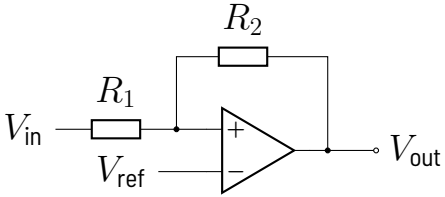
- op-amps are slow when used open loop: recovery from saturation, slew rate (µs)
- comparator ICs (LM393, LM339, TLV3501): fast, no frequency compensation, logic outputs
  - LM393/LM339: **open-collector** output, needs a pull-up resistor (e.g. $4.7\,\mathrm{k\Omega}$ to $5\,\mathrm{V}$)
- input offset voltage (mV): shifts the threshold; must be $\ll$ LSB
- input bias current: flows through the ladder resistors, small threshold shift
- never use a comparator IC as an op-amp: it has no compensation and oscillates with feedback

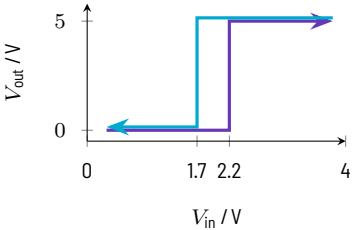
## Chatter at slow, noisy inputs

- a slowly changing input with noise crosses the threshold several times
- the output switches back and forth (**chatter**): a counter counts wrong, an LED flickers
- remedy: **hysteresis**: two thresholds, $V_{T+}$ for rising and $V_{T-}$ for falling inputs
  - positive feedback (Schmitt trigger)
  - the hysteresis $V_{T+} - V_{T-}$ must be larger than the peak-to-peak noise
- Falstad: triangle $0$ V … $4$ V at $1$ Hz plus $0.2$ V at $50$ Hz
  - plain comparator: 10 edges in one period; Schmitt trigger: exactly 2

▶ Falstad: `examples/adc/comparator_hyst.txt`

# Schmitt trigger (non-inverting)




- switching when $V_+ = V_{ref}$, with $V_+ = V_{in}\frac{R_2}{R_1+R_2} + V_{out}\frac{R_1}{R_1+R_2}$:

  $$V_T = V_{ref}\left(1 + \frac{R_1}{R_2}\right) - V_{out}\frac{R_1}{R_2}$$

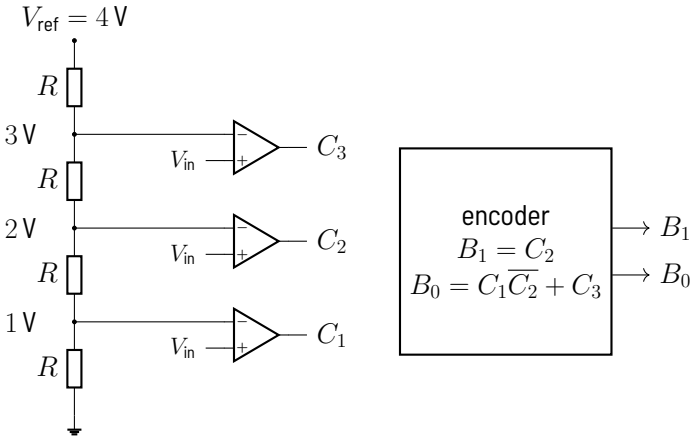
- $V_{T+}$ (output low): $V_{ref}(1 + R_1/R_2)$; $V_{T-}$ (output high): $V_{T+} - V_{OH}R_1/R_2$
- $R_1 = 10\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$, $V_{ref} = 2\,\text{V}$: $V_{T+} = 2.2\,\text{V}$, $V_{T-} = 1.7\,\text{V}$, hysteresis $0.5\,\text{V}$

## Design card: Comparator

- $V_{\text{out}} =$ high if $V_+ > V_-$, else low; op-amp open loop or comparator IC
- logic compatible: supply $0/5$ V, rail-to-rail output (or open collector + pull-up $4.7\,\text{k}\Omega$)
- Falstad: op-amp, max output $5$ V, min output $0$ V; logic inputs switch at $2.5$ V
- offset voltage $\ll$ LSB; input current $\times$ source resistance $\ll$ LSB
- hysteresis against chatter: $R_1$ from $V_{\text{in}}$ to $V_+$, $R_2$ from $V_{\text{out}}$ to $V_+$, $V_{\text{ref}}$ at $V_-$
- $V_{T+} = V_{\text{ref}}(1 + R_1/R_2)$, $V_{T-} = V_{T+} - V_{\text{OH}} R_1/R_2$, hysteresis $V_{\text{OH}} R_1/R_2 >$ noise (peak-to-peak)
- in a flash ADC behind an S&H: no hysteresis needed (the held input is constant), hysteresis would shift the thresholds

# 9.4 The Flash Converter

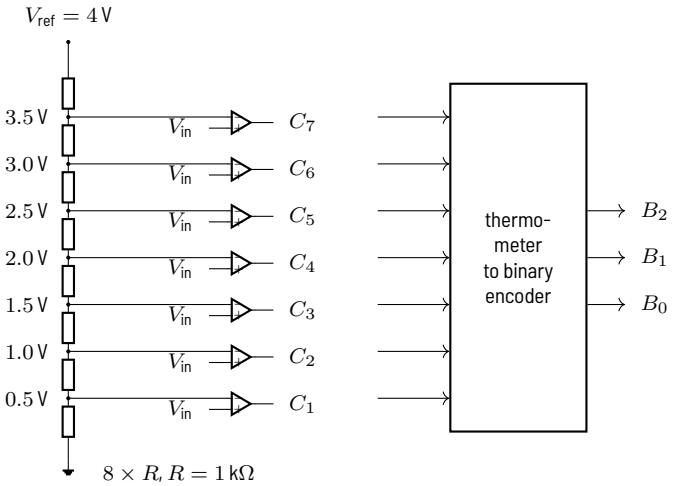
## 2-bit flash converter




- LSB $= 1\,\text{V}$; thresholds $1\,\text{V}$, $2\,\text{V}$, $3\,\text{V}$; 3 comparators
- $C_k = 1$ if $V_{\text{in}} > k \cdot \text{LSB}$

▶ Falstad: `examples/adc/flash2.txt`

# 3-bit flash converter (capstone)




- 7 comparators, $C_k = (V_{\text{in}} > k \cdot 0.5\,\text{V})$; ladder current $4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$

# Thermometer code

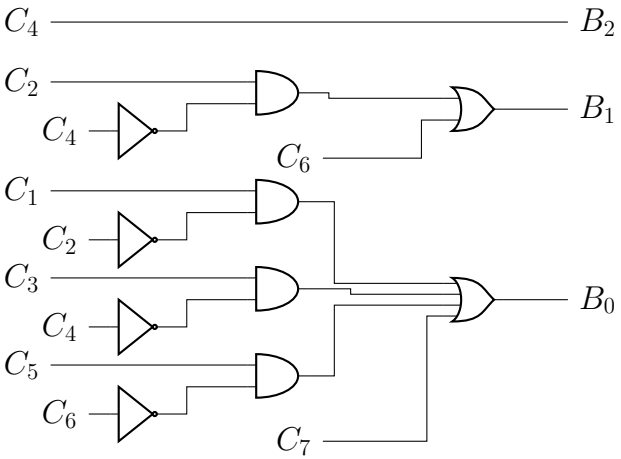
| $V_{\text{in}}$ / V | $D$ | $C_7$ | $C_6$ | $C_5$ | $C_4$ | $C_3$ | $C_2$ | $C_1$ | $B_2$ | $B_1$ | $B_0$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.00 ...0.50 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0.50 ...1.00 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1.00 ...1.50 | 2 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1.50 ...2.00 | 3 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 |
| 2.00 ...2.50 | 4 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 |
| 2.50 ...3.00 | 5 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 |
| 3.00 ...3.50 | 6 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 3.50 ...4.00 | 7 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

- like the column of a thermometer: all comparators below the input level are 1
- the number of ones is the code $D$; only $2^N$ of the $2^{2^N-1}$ patterns are valid

## Deriving the encoder

- the top of the column: $H_k = C_k \overline{C_{k+1}}$ is 1 only for $D = k$ (one-hot code, $C_8 := 0$)
- each output bit is the OR of the $H_k$ whose code has this bit set:
  - $B_2 = H_4 + H_5 + H_6 + H_7 = C_4$
  - $B_1 = H_2 + H_3 + H_6 + H_7 = C_2\overline{C_3} + C_3\overline{C_4} + C_6\overline{C_7} + C_7 = C_2\overline{C_4} + C_6$
  - $B_0 = H_1 + H_3 + H_5 + H_7 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$
- simplification uses the thermometer property: $C_{k+1} = 1 \Rightarrow C_k = 1$
  - $C_2\overline{C_3} + C_3\overline{C_4} = C_2\overline{C_4}$: “between level 2 and level 4”
  - $B_2$: four ones in a row means $D \geq 4$

# Encoder logic




- 3 NOT, 4 AND, 2 OR gates: the same gates as in the Falstad model

# Falstad: 3-bit flash converter with display

- input: slider $0$ V … $4$ V (label `Vin`); ladder $8 \times 1$ k$\Omega$, $4$ V
- comparator outputs and encoder signals are labeled nodes `C1`…`C7`, `nC2`, `a24`, …, `B0`…`B2`
- 7 LEDs as bar graph, 3 LEDs for $B_2 B_1 B_0$, each with $330\,\Omega$
- checked at the bin centres $0.25$ V, $0.75$ V, …, $3.75$ V: codes 000, 001, …, 111
- default $2.6$ V: five bar LEDs on, binary 101

▶ Falstad: `examples/adc/flash3.txt`

## Priority encoder and bubble errors

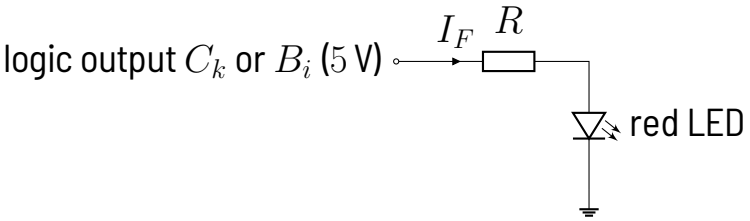
- **bubble**: a wrong comparator decision inside the column, e.g. by offset or noise near a threshold
  - true code 3: 0000111; with bubble at $C_2$: 0000101
  - our encoder: $B_0 = C_1\overline{C_2} = 1$, $B_1 = 0$, $B_2 = 0 \Rightarrow$ 001: error of 2 LSB
- **priority encoder** (74HC148, 8 inputs $\rightarrow$ 3 bit): outputs the number of the **highest** active input
  - ignores everything below the top: a bubble does no harm
- bubble correction with 3-input gates: $H_k = C_{k-1}\, C_k\, \overline{C_{k+1}}$
- large flash converters: ones counter (Wallace tree) or Gray-code encoding

## Design card: Thermometer-to-Binary Encoder

- thermometer code: $C_k = 1$ for all $k \leq D$; the number of ones is the code $D$
- one-hot: $H_k = C_k \overline{C_{k+1}}$ (top of the column); $B_i =$ OR of all $H_k$ with bit $i$ of $k$ set
- 3 bit (capstone): $B_2 = C_4 \quad B_1 = C_2\overline{C_4} + C_6 \quad B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$
- gates: 3 NOT ($\overline{C_2}, \overline{C_4}, \overline{C_6}$), 4 AND2, 1 OR2, 1 OR4
- check: sweep $V_{\text{in}}$ over the bin centres $0.25$ V … $3.75$ V in steps of $0.5$ V: 000 …111
- alternative: priority encoder (74HC148, active-low inputs and outputs), robust against bubbles

# 9.5 Displaying the Result

## LED with series resistor




**Series resistor**

$$R = \frac{V_{\text{OH}} - V_F}{I_F} = \frac{5\,\text{V} - 2\,\text{V}}{9\,\text{mA}} = 333\,\Omega \quad \rightarrow \quad \text{E12: } 330\,\Omega,\ I_F = 9.1\,\text{mA}$$

- red LED: $V_F \approx 1.8\,\text{V} \ldots 2.0\,\text{V}$; bright enough at $5\,\text{mA} \ldots 10\,\text{mA}$
- **one resistor per LED**: with a common resistor the current would depend on the number of LEDs that are on
- Falstad LED model: $V_F = 1.77\,\text{V}$, $I_F = 9.8\,\text{mA}$ with $330\,\Omega$

## Current budget

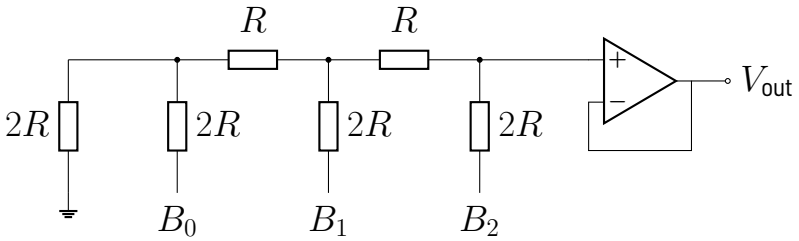
- code 7: 7 bar LEDs + 3 binary LEDs on: $10 \times 9.1\,\text{mA} \approx 91\,\text{mA}$ from the $5\,\text{V}$ supply
- each output must deliver its LED current at a high level close to $5\,\text{V}$
  - in our circuit $C_4$ drives its bar LED, the $B_2$ LED and the gate inputs: $18\,\text{mA}$
  - 74HC gates: $\pm 25\,\text{mA}$ absolute maximum per output, $V_{\text{OH}}$ drops under load
- better: buffer gates, low-current LEDs ($2\,\text{mA}$), or a driver IC (74HC245; ULN2003 sinks)
- LED to ground (output sources current) or LED to $5\,\text{V}$ (output sinks, active low): both are fine

## Design card: Falstad digital cheat-sheet

- gates: **Draw** $\rightarrow$ **Logic Gates, Input and Output**: AND, OR, NAND, NOR, XOR, Inverter; number of inputs and high level ($5$ V) in the edit dialog (double click)
- logic levels: input high above $V_{\text{high}}/2 = 2.5$ V; outputs are ideal $0$ V/$5$ V sources
- comparator: op-amp with max output $5$ V and min output $0$ V; inputs $-$ and $+$ as drawn
- **Logic Input** (click to toggle) and **Logic Output** (shows H/L) for quick tests
- **Labeled Node**: nodes with the same name are connected; use them for $C_1 \ldots C_7$, $B_0 \ldots B_2$; hover to read the voltage
- input voltage: DC voltage source or rail with a slider (edit dialog); clock: rail with square wave, frequency, duty cycle
- analog switch: closed if control $> 2.5$ V, $R_{\text{on}} = 20\,\Omega$
- LED: always with a series resistor ($330\,\Omega$ at $5$ V); lights above a few mA
- scope: right click on a node or element $\rightarrow$ **View in Scope**; set the time step to $0.1$ ms for $5$ Hz clocks

# 9.6 Other Converter Architectures

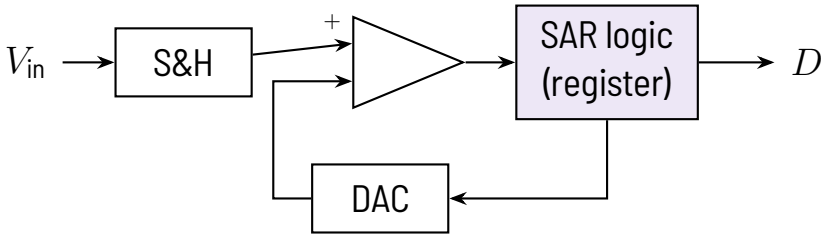
## R-2R ladder DAC




- bit inputs: logic levels $0\,\text{V}$ or $V_{\text{ref}}$; output resistance of the ladder: always $R$
- $V_{\text{out}} = V_{\text{ref}} \left( \frac{B_2}{2} + \frac{B_1}{4} + \frac{B_0}{8} \right) = V_{\text{ref}} \frac{D}{2^N}$
- only two resistor values, any number of bits: easy to integrate
- Falstad: $R = 10\,\text{k}\Omega$, $5\,\text{V}$ logic, counter at $1\,\text{Hz}/0.5\,\text{Hz}/0.25\,\text{Hz}$: staircase in steps of $0.625\,\text{V}$, $D = 5$: $3.125\,\text{V}$

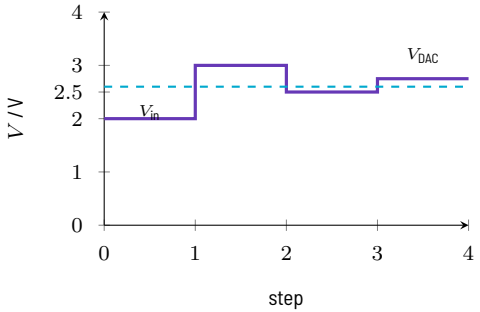
▶ Falstad: `examples/adc/r2r_dac.txt`

# Successive approximation (SAR)




- binary search with one comparator and a DAC: one bit per clock cycle, MSB first
- set the bit, compare $V_{\text{in}}$ with $V_{\text{DAC}}$: keep the bit if $V_{\text{in}} \geq V_{\text{DAC}}$, otherwise clear it
- $N$ steps for $N$ bits; needs an S&H (built in); 8 ...18 bit, up to some MS/s

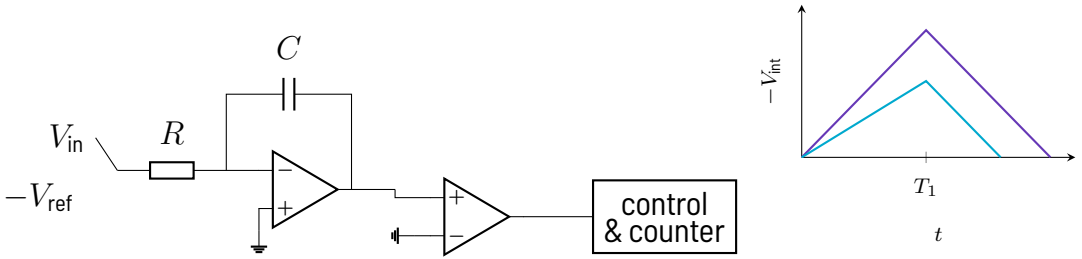
# SAR: worked example




| step | trial | $V_{DAC}$ | $V_{in} \geq V_{DAC}$? |
|---|---|---|---|
| 1 | **1**000 | $2.00\,V$ | yes: keep |
| 2 | 1**1**00 | $3.00\,V$ | no: clear |
| 3 | 10**1**0 | $2.50\,V$ | yes: keep |
| 4 | 101**1** | $2.75\,V$ | no: clear |
| | **1010** | $2.50\,V$ | $D = 10$ |

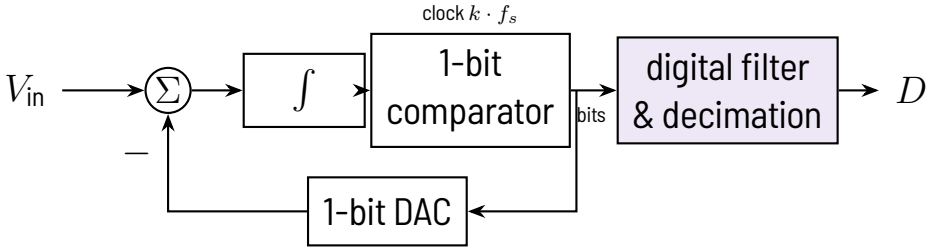
- 4 bit, $V_{FS} = 4\,V$, LSB $= 0.25\,V$, $V_{in} = 2.6\,V$
- result 1010: $V_{in}$ lies between $2.50\,V$ and $2.75\,V$

# Dual-slope converter




- integrate $V_{in}$ for a fixed time $T_1$, then integrate $-V_{ref}$ until the output is zero again ($T_2$)
- $V_{in}\,T_1 = V_{ref}\,T_2 \quad \Rightarrow \quad D = N_1 \dfrac{V_{in}}{V_{ref}}$: independent of $R$, $C$ and the clock frequency
- $T_1 = 20$ ms (or $100$ ms): 50 Hz interference integrates to zero

# Sigma-delta converter (concept)




- oversampling with a 1-bit quantizer in a feedback loop: the density of ones is proportional to $V_{\text{in}}$
- the loop shapes the quantization noise to high frequencies; a digital low-pass removes it
- 16 ...24 bit, audio and precision measurement (weighing scales, thermocouples)
- the digital filter is also the anti-aliasing filter: the analog filter can be simple

# Comparison

| | flash | SAR | dual-slope | sigma-delta |
|---|---|---|---|---|
| principle | $2^N - 1$ comparators | binary search | integration | oversampling |
| steps per sample | 1 | $N$ | $2 \cdot 2^N$ clocks | many (oversampling) |
| resolution | 4 …8 bit | 8 …18 bit | 12 …22 bit | 16 …24 bit |
| speed | up to GS/s | kS/s …MS/s | 1 …100 S/s | Hz …MHz bandwidth |
| cost, power | high (exponential) | low | low | medium |
| typical use | oscilloscopes, radio | microcontrollers | multimeters | audio, scales |

- the capstone uses a 3-bit flash converter because every part of it is visible: ladder, comparators, logic

# 9.7 Exercises

## Exercise 9.1: Aliasing

A system samples at $f_s = 5$ Hz.

a) Calculate the alias frequencies of input components at $2$ Hz, $4$ Hz, $12$ Hz, $49$ Hz and $50$ Hz.

b) The sensor signal is $2$ V DC with $0.5$ V of $50$ Hz hum. Describe the sampled values for $f_s = 5$ Hz and for $f_s = 4.9$ Hz.

c) An RC low-pass with $6.8$ kΩ and $10$ µF is inserted before the S&H stage. Calculate $f_c$ and the remaining hum amplitude. Compare with the LSB of the 3-bit converter ($0.5$ V).

d) Check part b) in Falstad.

# Solution 9.1

a) $2\,\text{Hz}$: below $f_s/2$, no alias. $4\,\text{Hz}$: $|4-5| = 1\,\text{Hz}$. $12\,\text{Hz}$: $|12 - 2\cdot 5| = 2\,\text{Hz}$. $49\,\text{Hz}$: $|49 - 10\cdot 5| = 1\,\text{Hz}$. $50\,\text{Hz}$: $|50 - 10\cdot 5| = 0\,\text{Hz}$.

b) $f_s = 5\,\text{Hz}$: the hum is sampled at the same phase every time; all samples are equal, $2\,\text{V} + 0.5\,\text{V}\sin\varphi_0$, i.e. a constant offset between $-0.5\,\text{V}$ and $0.5\,\text{V}$. $f_s = 4.9\,\text{Hz}$: the phase advances by $0.204$ periods per sample; the samples follow a $1\,\text{Hz}$ sine between $1.5\,\text{V}$ and $2.5\,\text{V}$.

c) $f_c = 1/(2\pi\cdot 6.8\,\text{k}\Omega\cdot 10\,\mu\text{F}) = 2.34\,\text{Hz}$. $|H(50\,\text{Hz})| = 1/\sqrt{1+(50/2.34)^2} = 0.047$ ($-26.6\,\text{dB}$): the hum is reduced to $23\,\text{mV}$, less than $\frac{1}{20}$ LSB. The offset error disappears from the readings.

d) ▶ Falstad: `examples/adc/alias50.txt`

`Vs_5Hz` holds $2.14\,\text{V}$ at every sample (offset $0.14\,\text{V}$ for the phase in the example); `Vs_4.9Hz` shows $2.14\,\text{V}$, $2.50\,\text{V}$, $2.14\,\text{V}$, $1.58\,\text{V}$, $1.62\,\text{V}$, $2.20\,\text{V}$, ...

## Exercise 9.2: Quantization

A 3-bit converter has a full scale of $4$ V and thresholds at $k \cdot$ LSB.

a) Calculate the LSB and the codes for $0.3$ V, $1.8$ V, $2.0$ V and $3.98$ V.
b) Which voltage range does the code 101 represent? What is the largest quantization error?
c) Calculate the SNR of the ideal 3-bit converter. How many bits are needed for an SNR of at least $60$ dB?
d) Observe the quantization in Falstad.

# Solution 9.2

a) LSB $= 0.5\,\text{V}$. $D = \lfloor V_\text{in}/0.5\,\text{V} \rfloor$: $0.3\,\text{V} \to$ 000, $1.8\,\text{V} \to$ 011, $2.0\,\text{V} \to$ 011 or 100 (exactly on the threshold, the comparator offset decides), $3.98\,\text{V} \to$ 111.

b) 101 = 5: $2.5\,\text{V} \ldots 3.0\,\text{V}$. The error $V_\text{in} - D \cdot \text{LSB}$ is between 0 and $0.5\,\text{V}$ (1 LSB); relative to the bin centre $2.75\,\text{V}$ it is $\pm 0.25\,\text{V}$.

c) SNR $= 6.02 \cdot 3 + 1.76 = 19.8\,\text{dB}$. $6.02\,N + 1.76 \geq 60 \Rightarrow N \geq 9.7$: 10 bits.

d) ▶ Falstad: `examples/adc/quant_error.txt`

At $V_\text{in} = 1.8\,\text{V}$ the DAC output is $V_q = 1.5\,\text{V}$, at $3.98\,\text{V}$ it is $3.5\,\text{V}$; $V_\text{in} - V_q$ stays between $0\,\text{V}$ and $0.5\,\text{V}$.

## Exercise 9.3: Designing the 3-bit flash converter

Design the converter of the capstone: full scale $4.00\,\text{V}$, 3 bit, comparators $C_k = (V_\text{in} > k \cdot 0.5\,\text{V})$.

a) Dimension the reference ladder with $1\,\text{k}\Omega$ resistors. Calculate the tap voltages, the ladder current and the power taken from the reference.

b) The comparators have an input bias current of $100\,\text{nA}$. Estimate the worst-case threshold shift. (Hint: Thevenin resistance of a tap.)

c) Write down the truth table from thermometer code to binary and derive the logic equations for $B_2, B_1, B_0$.

d) How must the ladder be changed for thresholds at $(k - \tfrac{1}{2}) \cdot \text{LSB}$?

e) Check the converter in Falstad at the bin centres.

# Solution 9.3

a) 8 resistors of $1\,\mathrm{k\Omega}$ from $4\,\mathrm{V}$ to ground. Taps $V_k = k{\cdot}4\,\mathrm{V}/8 = 0.5\,\mathrm{V}, 1.0\,\mathrm{V}, \ldots, 3.5\,\mathrm{V}$. $I = 4\,\mathrm{V}/8\,\mathrm{k\Omega} = 0.5\,\mathrm{mA}$, $P = 2\,\mathrm{mW}$.

b) The largest Thevenin resistance is at the middle tap: $4\,\mathrm{k\Omega} \parallel 4\,\mathrm{k\Omega} = 2\,\mathrm{k\Omega}$. Shift: $100\,\mathrm{nA} \cdot 2\,\mathrm{k\Omega} = 0.2\,\mathrm{mV}$, negligible compared with $0.5\,\mathrm{V}$.

c) Truth table as on the slide **Thermometer code**. With $H_k = C_k\overline{C_{k+1}}$: $B_2 = H_4 + H_5 + H_6 + H_7 = C_4$; $B_1 = H_2 + H_3 + H_6 + H_7 = C_2\overline{C_4} + C_6$; $B_0 = H_1 + H_3 + H_5 + H_7 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$.

d) Total resistance stays $8R$: bottom resistor $R/2 = 500\,\Omega$, six resistors $R$, top resistor $3R/2 = 1.5\,\mathrm{k\Omega}$. The taps are then $0.25\,\mathrm{V}$, $0.75\,\mathrm{V}$, ..., $3.25\,\mathrm{V}$.

e) ▶ Falstad: `examples/adc/flash3.txt`

Taps $0.5\,\mathrm{V}$ ...$3.5\,\mathrm{V}$, ladder current $0.50\,\mathrm{mA}$. Slider at $0.25\,\mathrm{V}$, $0.75\,\mathrm{V}$, ..., $3.75\,\mathrm{V}$: the bar graph shows 0 ...7 LEDs, the binary LEDs 000, 001, ..., 111.

## Exercise 9.4: Bubble error

Because of noise, the comparators of the 3-bit flash converter deliver the pattern 0010111 ($C_7$ first), a bubble at $C_4$.

a) Which input range is plausible? Which code would be correct?
b) Which code does the encoder of the capstone produce?
c) Which code does a priority encoder produce?

# Solution 9.4

a) $C_5 = 1$ but $C_4 = 0$ contradict each other. Either $C_4$ is wrong (input between $2.5$ V and $3.0$ V, code 5) or $C_5$ is wrong (input between $1.5$ V and $2.0$ V, code 3).

b) $B_2 = C_4 = 0$; $B_1 = C_2\overline{C_4} + C_6 = 1$; $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7 = 0 + 1 + 1 + 0 = 1$: code 011 = 3, one of the plausible values. (For the pattern 0000101 of the slide, however, it gives 001 instead of 011.)

c) The priority encoder looks at the highest active input, $C_5$: code 101 = 5, the other plausible value. For 0000101 it gives 011 = 3, the correct value; it never produces a code below the top of the column.

## Exercise 9.5: LED display

The bar graph and the binary LEDs are driven by $5$ V logic outputs. Red LEDs with $V_F \approx 2$ V shall carry $5$ mA … $10$ mA.

a) Choose the series resistor from the E12 series and calculate the LED current.
b) Calculate the total LED current for the code 111.
c) In the Falstad model, $C_4$ drives its bar LED and also the $B_2$ LED. Which current does this output deliver?
d) Why must each LED have its own resistor?

# Solution 9.5

a) $R = (5\,\text{V} - 2\,\text{V})/I_F$ gives $300\,\Omega$ ...$600\,\Omega$. Choose $330\,\Omega$: $I_F = 3\,\text{V}/330\,\Omega = 9.1\,\text{mA}$.

▶ Falstad: `examples/adc/flash3.txt`

The Falstad LED has $V_F = 1.77\,\text{V}$, so the current is $3.23\,\text{V}/330\,\Omega = 9.8\,\text{mA}$.

b) 10 LEDs: $10 \cdot 9.1\,\text{mA} = 91\,\text{mA}$.

c) $2 \cdot 9.1\,\text{mA} = 18\,\text{mA}$ (plus negligible gate input currents). That is near the limit of a CMOS logic output; a buffer or lower LED currents ($680\,\Omega$: $4.4\,\text{mA}$) are safer.

d) LEDs of different colours and samples have different $V_F$. With a common resistor the LED with the lowest $V_F$ takes most of the current, and the current per LED changes with the number of LEDs that are on.

## Exercise 9.6: Schmitt trigger, R-2R DAC and SAR

a) A non-inverting Schmitt trigger has $V_{\text{ref}} = 2\,\text{V}$, $R_1 = 10\,\text{k}\Omega$ and output levels $0\,\text{V}/5\,\text{V}$. Choose $R_2$ for a hysteresis of $0.5\,\text{V}$ and calculate both thresholds.

b) A 3-bit R-2R DAC is driven by $5\,\text{V}$ logic levels. Calculate the step size and the output for $D = 5$.

c) A 4-bit SAR converter ($V_{\text{FS}} = 4\,\text{V}$) converts $V_{\text{in}} = 1.3\,\text{V}$. Write down the trial codes, the DAC voltages and the result.

d) Check a) and b) in Falstad.

# Solution 9.6

a) Hysteresis $V_{0H} R_1/R_2 = 0.5\,\text{V} \Rightarrow R_2 = 5\,\text{V} \cdot 10\,\text{k}\Omega/0.5\,\text{V} = 100\,\text{k}\Omega$. $V_{T+} = 2\,\text{V} \cdot 1.1 = 2.2\,\text{V}$, $V_{T-} = 1.7\,\text{V}$.

b) LSB $= 5\,\text{V}/8 = 0.625\,\text{V}$; $D = 5$: $3.125\,\text{V}$.

c) LSB $= 0.25\,\text{V}$. 1000: $2.00\,\text{V}$, no $\rightarrow$ 0. 0100: $1.00\,\text{V}$, yes $\rightarrow$ 1. 0110: $1.50\,\text{V}$, no $\rightarrow$ 0. 0101: $1.25\,\text{V}$, yes $\rightarrow$ 1. Result 0101 = 5 ($1.25\,\text{V} \ldots 1.50\,\text{V}$).

d) ▶ Falstad: `examples/adc/comparator_hyst.txt`

The Schmitt trigger switches at $2.2\,\text{V}$ (rising) and $1.7\,\text{V}$ (falling) and gives exactly two edges per period; the plain comparator produces 10 edges.

▶ Falstad: `examples/adc/r2r_dac.txt`

`Vout` steps through $0\,\text{V}$, $0.625\,\text{V}$, ..., $4.375\,\text{V}$; at $D = 5$ it shows $3.125\,\text{V}$.